

AIE1901: Large Language Models for Social Simulations

Mid-term Group Presentation — Option A

School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen

Deliverables

Submit: Submit a single ZIP named: `AIE1901_Midterm_GroupX_OptionA.zip`

Deliverables inside ZIP:

- `slides.pdf` or `slides.pptx`
- `video.mp4` or a link in `README.txt`
- (optional) additional data or plots

Purpose

This mid-term presentation is a comprehensive documentation and presentation of your progress in social simulation modeling, integrating your work from Lab 1–3.

It focuses on your analysis and understanding of the simulations from the following two papers:

- *Giant catalytic effect of altruists in Schelling’s segregation model*
- *Competition between collective and individual dynamics*

Your goal is to clearly demonstrate your understanding of:

- The city model and its variations.
- The parameter studies you conducted.
- The key insights you derived.

Your Presentation Should Cover

Each group (Option A) should prepare a comprehensive slide deck (no page limit) addressing the following key components:

1. Overview

- Introduce the background and motivation of the two papers.
- Explain the main objectives: how the authors simulate social behavior and collective dynamics via agent-based models.

2. Methodology and System Architecture

- Describe the architecture of modeling framework in the two papers.
- Show diagrams (either from the papers or your own) illustrating the system flow.

3. Code Structure and Implementation Analysis

- Provide an overview of the code repository or scripts used in your Labs.
- Explain the purpose of each major module or file.
- Discuss how these modules work together during simulations.

4. Parameter Studies

- Compare distinct scenarios or model setups.
- Highlight how changing key parameters influenced simulation outcomes.
- Show representative plots, heatmaps, or diagrams to illustrate trends.
- Discuss your observation patterns: When does segregation increase or decrease? What behaviors emerge? Which parameters are most influential?

5. Insights & Reflections

- What did you learn from studying or experimenting with the two simulation systems presented in the papers?
- What limitations, challenges, or open questions did you identify through your analysis?
- How might similar architectures evolve in future social simulation studies?

Short Video (≤ 3 minutes)

In addition to the slides, produce a short video summary. This can take one of two forms:

- **Mini Presentation:** A concise talk presenting the highlights of your slides.
- **Edited Summary Video:** A short, well-edited video combining simulation visuals, narration, and key findings.

Requirements:

- Length: ≤ 3 minutes.
- Must contain substantive content (not just general introductions).

Assessment Rubric (100 pts)

Criterion	Description	Points
Conceptual Integration	Clear linkage of contents into a cohesive story	25
Parameter Study & Scenario Analysis	Depth, correctness, and insightfulness of your experiments	20
Presentation Quality	Clarity, organization, and use of visuals	20
Video Content	Conciseness, informativeness, and engagement	20
Professionalism	File naming and formatting	15